

informational services aimed at education and community development in rural India. The aim of the project is to create an IT-based rural information service so that the population has access to critical information via the Internet, such as updates on farming practices and hygiene. In view of the fact that Andhra Pradesh (AP) has a total of 7,500 predominantly agriculture-based villages, the task at hand was monumental.

YonEarth interactive, a Mumbai based IT-enabled marketing solutions company, was hired to design, develop and manage the venture.

"Every one knew it was an extremely uphill task," says Vivek Marolli, Director, YonEarth Interactive. "It was completely untreaded territory. Making such an ambitious project work in far-flung villages across rural India, under difficult conditions such as sporadic power supply, heat, dust, poor computer literacy and almost non-existent or very slow Internet connectivity, would indeed be a challenge. We had to design a system that would beat all these barriers and work in ways that were never thought possible before."

After a study of the requirements for the project, and a year spent on designing and building the software, the system was ready to be launched. While building the ICT—the Integrated Circuit Technology—the aim was to have a villagers get their hands

on a computer rather than employ a kiosk owner as the interface between the technology and the user. Hence, the project emphasised on using a voice- and video-based system, and uses "rich media"—software designed using Macromedia Flash—besides being database-driven, so that all the intelligence can be built on it.

A voice system would eliminate the necessity for the user to read off the screen, hence making it easier for an illiterate user. It was also important to have a system that would allow the central servers to connect to these kiosks through the barriers of poor Internet connectivity, and enable large volumes of rich-media files and information to go back and forth between the Mumbai office and thousands of such kiosks on an everyday basis. Finally, the users had to be taught to use and understand what they stood to gain by using the system, besides being trained to use the mouse and keyboard.

Each i-Shakti kiosk is owned by an individual, and it comprises a desktop PC system that operates independently of all other kiosks on the network, but is connected to the central server to which it can synchronise data through a dial-up connection. The i-Shakti software needs to be installed on PCs with suitable configurations.

Once deployed on the PC, it acts as a kiosk, and the owner of the kiosk is expected to synchronise the data and transfer it to the central server once a day. The kiosk owner also helps new users



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Director
YonEarth
Interactive



register and guides them when they use the system. The central server communicates to thousands of such kiosks through a specially-developed Compressed Data Synchronization-based Dialogue Technology, which allows the kiosks to work in offline mode. This reduces the cost of operations to the kiosk owner, and at the same time, he can connect to the Central Server every day for two-way data updates. Technical support teams, which have been set up all over AP, also monitor and solve any problems on the kiosks within the state. For every forty kiosks, there is a technical manager who tackles any problems that the kiosks face.

Currently, every one of the i-Shakti kiosks in AP has various channels of interactive rich-media information created by culling various sources such as opinions of agriculture experts, the Internet, and government agencies, on topics such as health, agriculture, animal husbandry, education, employment, women's issues, and entertainment. As a registered user, one can visit the i-Shakti kiosk in his village six days a week and access information and services. Since there are experts who are a part of the project—some on a payroll and some voluntary—the villagers can also post questions on a query mailing system.



Why i-Shakti?

How is such information critical to the villagers? Well, most villages are isolated and independently-functioning entities, and they rely on their existing means of information and resources to aid their everyday activities. This implies that in case of an illness, they have access only to a local doctor, and possibly no chance at a second opinion. With i-Shakti, the villager has access to information that comes to him from a recognised government source, and is an authorised voice—most of the experts on the panel of i-Shakti are people employed with various agencies all over AP. For example, the expert on wheat might be an agricultural scientist at the University of Vijaywada. Besides, the information is free, and one can gain insights in any developments in



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